

BASICS & VARIABLES

```
x = 5          # integer
y = 3.14       # float
name = "Alice" # string
ok = True      # boolean
z = None       # null value
a, b, c = 1, 2, 3
```

DATA TYPES

```
int      10
float    3.14
str       "text"
bool     True / False
list     [1, 2, 3]
tuple    (1, 2, 3)
dict     {"key": "value"}
set      {1, 2, 3}
```

OPERATORS

```
+ - * / // (floor div)
% ** (power)
== != > < >= <=
and or not
```

STRINGS

```
s = "Hello World"
s.lower()      # hello world
s.upper()      # HELLO WORLD
s.strip()      # trim whitespace
s.split(" ")   # ['Hello', 'World']
s.replace("H", "J") # Jello World
len(s)         # 11
s[0]           # 'H'
s[-1]          # 'd'
s[0:5]         # 'Hello'
f"Name: {name}" # f-string
```

LISTS

```
lst = [1, 2, 3]
lst.append(4)    # add item
lst.remove(2)    # remove value
lst.pop()        # remove last
lst.sort()       # sort ascending
lst.reverse()    # reverse order
len(lst)         # length
lst[0]           # access item
lst[1:3]         # slicing
```

DICTIONARIES

```
d = {"name": "Alice", "age": 25}
d["name"]        # get value
d.get("age")     # safe get
d.keys()         # all keys
d.values()       # all values
d.items()        # key-value pairs
d["city"] = "NYC" # add/update
del d["age"]     # remove key
```

TUPLES & SETS

```
t = (1, 2, 3)    # immutable
t[0]             # 1

s = {1, 2, 3}
s.add(4)
s.remove(1)
s1 | s2          # union
s1 & s2          # intersection
```

TYPE CONVERSION

```
int("5")        # 5
float("3.1")     # 3.1
str(5)           # "5"
list("abc")      # ['a', 'b', 'c']
bool(0)          # False
```

CONDITIONALS

```
if x > 10:
    print("big")
elif x == 10:
    print("equal")
else:
    print("small")
```

LOOPS

```
for i in range(5):
    print(i)

for item in [1, 2, 3]:
    print(item)

while x < 10:
    x += 1

break      # exit loop
continue  # skip iteration
```

FUNCTIONS

```
def greet(name, greeting="Hi"):
    return f"{greeting}, {name}!"

greet("Sam")      # Hi, Sam!
greet("Sam", "Yo") # Yo, Sam!

square = lambda x: x ** 2
```

LIST COMPREHENSIONS

```
squares = [x**2 for x in range(10)]
evens = [x for x in range(10)
         if x % 2 == 0]
```

ERROR HANDLING

```
try:
    x = 1 / 0
except ZeroDivisionError:
    print("Can't divide by zero")
finally:
    print("Done")
```

CLASSES

```
class Dog:
    def __init__(self, name):
        self.name = name

    def bark(self):
        return f"{self.name} says Woof!"

d = Dog("Rex")
d.bark()
```

USEFUL BUILT-INS

```
len()    type()    range()
print()  input()   sorted()
min()    max()     sum()
enumerate() zip()  map()
filter() abs()     round()
```

IMPORT & FILES

```
import math
from datetime import datetime

with open("file.txt", "r") as f:
    content = f.read()

with open("out.txt", "w") as f:
    f.write("hello")
```

COMMON PATTERNS

```
# swap variables
a, b = b, a

# ternary
y = "even" if x % 2 == 0 else "odd"

# unpack with star
first, *rest = [1, 2, 3, 4]

# dict comprehension
sq = {x: x**2 for x in range(5)}
```

STRING FORMATTING

```
f"{3.14159:.2f}" # '3.14'
f"{42:05d}"      # '00042'
f"{name!r}"      # repr
("{} and {}".format(1, 2))
```